
Do World Action Models Use Their Predicted Futures?

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Abstract

World action models generate predicted future observations together with robot actions, but it remains unclear whether those futures influence action generation. We introduce future transplantation to test this directly. From the same simulator state, we sample two continuations and recompute one after replacing its future representation with the other’s, while holding the current inputs and sampling noise fixed. We then ask whether the recomputed action and its one-chunk simulator endpoint move toward the behavior paired with the donor future. In Cosmos Policy, at the first policy query in each of ten LIBERO-10 tasks, donor futures increase action projection by 0.499 and endpoint projection by 0.552 relative to self-future recomputation. Effects are much smaller at later sampled states. Camera-future replacements also steer actions, whereas future proprioception and isolated object-state changes have near-zero effects. This pattern is consistent with sensitivity to camera-visible robot motion. In Cosmos 3, transplanted futures steer actions and endpoints across four RoboLab states. Replacing future-token keys and values wherever action tokens directly attend to future tokens reduces the action shift by 0.842 projection units on four calibration states and substantially attenuates it in one held-out state. These results show that generated future content can influence WAM actions. They do not establish that either model compares predicted task outcomes or performs planning.

1 Introduction

World action models (WAMs) couple prediction and control: the same model generates possible future observations and the robot action to take next [8, 14]. This design suggests that a WAM may use its predicted future when producing an action. Joint generation alone, however, does not establish that computation. A future may be an unused output, a generic computational workspace, a visual description of the robot motion paired with the action, or a prediction of task consequences used to choose among actions.

These possibilities imply different accounts of the policy. If predicted robot motion drives the action, the model resembles visual inverse dynamics: it maps a current observation and a displayed future pose to the motion connecting them [1, 7]. If predicted object changes drive action selection, the computation is closer to visual planning, where candidate actions or subgoals are evaluated through their predicted outcomes [2, 3]. We study Cosmos Policy in its direct-policy mode and the diffusion-generator component of Cosmos3-Nano-Policy-DROID. Neither inference procedure evaluated here generates and scores multiple candidate actions before acting.

RIFT provides the closest causal evidence that future representations matter [21]. It masks cached values at future positions or assigns them to different positions, changing closed-loop trajectories and task success across several WAMs. These interventions establish dependence on future-position activations, but they cannot identify the information being used. An arbitrary corruption may move an action in any direction, even if it destroys no meaningful representation of an alternative behavior.

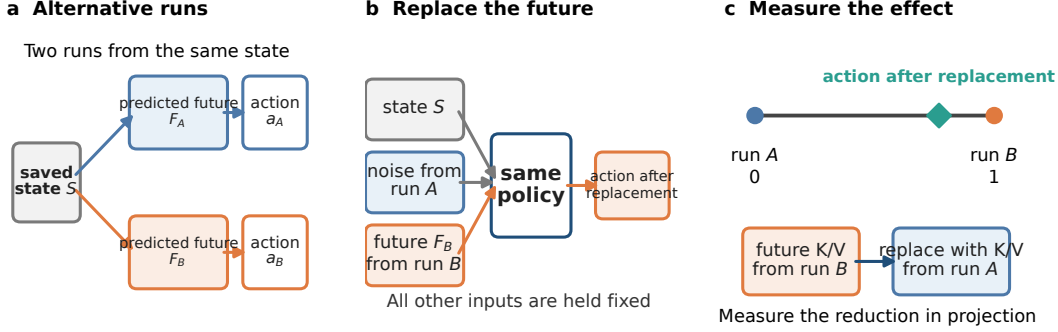


Figure 1: Future transplantation. We generate two continuations from one simulator state, recompute the recipient with the donor future, and project the new action and one-chunk endpoint onto the recipient-to-donor direction. In Cosmos 3, we additionally replace future-token keys and values to test their contribution to the action shift.

39 We introduce *reachable-donor future transplantation*. Starting from the same restored simulator
40 state, we sample two continuations from the unmodified policy. We designate one continuation as
41 the recipient and the other as the donor, then recompute the recipient with the donor’s predicted
42 future while keeping its current inputs and random draws fixed. The donor action and endpoint are
43 never given to the model; they are used only as references. We measure whether the recomputed
44 action and its executed endpoint move toward the donor behavior rather than merely away from the
45 recipient (Figure 1).

46 Transplanted futures steer actions and simulator endpoints toward the donor in both models. Cos-
47 mos Policy experiments further show much larger effects for camera futures than for future pro-
48 prioreception or isolated object-state changes. In Cosmos 3, replacing future-token keys and values
49 substantially attenuates the action shift. These findings answer three successively narrower ques-
50 tions: whether future content affects action, which changes in that content matter in Cosmos Policy,
51 and which attention activations contribute to the Cosmos 3 effect. Because the content and pathway
52 experiments use different models, they do not form a single end-to-end mechanism. They also do
53 not show that either model evaluates predicted task outcomes.

54 2 Related Work

55 Robot policies use predicted visual futures through several interfaces. Some methods first generate
56 a future video and then infer actions from that video or its representation [1, 7]. Others model video
57 and actions jointly [6, 8, 10, 14]. Joint training does not require that a model generate or use video at
58 inference time; UVA, for example, can omit video generation during inference [10]. Explicit visual
59 planners instead generate candidate actions or subgoals, predict their outcomes, and compare those
60 outcomes using a goal or value function [2, 3].

61 Recent WAM studies have begun to separate predictive training from the use of predictions during
62 action generation. Fast-WAM removes explicit future rollout while retaining competitive policy per-
63 formance [20]. Faster-WAM keeps a lower-cost connection from future representations to actions
64 and reports improved robustness in its evaluation settings [23]. AGRA finds that visually plausible
65 predictions do not necessarily support reliable action inference in one architecture [17]. RIFT di-
66 rectly intervenes on activations at future positions [21]. We instead replace one naturally generated
67 future with another and test whether the action follows the selected donor.

68 Future transplantation is an activation-replacement intervention. Such methods distinguish infor-
69 mation that is merely decodable from information that affects a model’s output [5, 12, 16, 22]. A
70 replacement can affect several computational routes at once, and untouched activations may com-
71 pensate for it [18]. We therefore use the Cosmos 3 intervention only to test whether future-token
72 keys and values contribute to the action shift, not to claim a complete causal decomposition.

73 3 Methods

74 3.1 Future transplantation and directional projection

75 Let S be a saved simulator state. Two runs of the unmodified policy from S produce branches A
 76 and B , with future representations F_A and F_B , action chunks $\mathbf{a}_A$ and $\mathbf{a}_B$, and endpoints $\mathbf{e}_A$ and $\mathbf{e}_B$
 77 obtained by executing one action chunk. We call A the recipient and B the donor.

78 To recompute branch A , we restore the same state and observation. We keep the instruction, action-
 79 noise realization, future-noise realization, solver, and denoising schedule fixed, but replace F_A with
 80 F_B . We also reverse the roles of A and B . For either an action chunk or endpoint, let $\mathbf{x} \in \{\mathbf{a}, \mathbf{e}\}$.
 81 We measure the projection of an intervened output $\hat{\mathbf{x}}$ onto the recipient-to-donor direction:

$$\pi_x(\hat{\mathbf{x}}; A, B) = \frac{(\hat{\mathbf{x}} - \mathbf{x}_A)^\top (\mathbf{x}_B - \mathbf{x}_A)}{\|\mathbf{x}_B - \mathbf{x}_A\|_2^2}. \quad (1)$$

82 The recipient has projection 0 and the donor has projection 1. We also recompute the recipient with
 83 its own future and define

$$\Delta_x = \pi_x(\hat{\mathbf{x}}_{\text{donor}}; A, B) - \pi_x(\hat{\mathbf{x}}_{\text{self}}; A, B). \quad (2)$$

84 A positive value means that the donor future moves the output toward the donor behavior. An
 85 arbitrary perturbation can change the output substantially while having zero or negative projection.

86 3.2 Models, states, and controls

87 For Cosmos Policy, we use the `Cosmos-Policy-LIBERO-Predict2-2B` checkpoint in direct-
 88 policy mode on the ten long-horizon LIBERO tasks, which we call LIBERO-10 [8, 11]. The pre-
 89 specified evaluation set contains 30 states: one per task at the first policy query, after three action
 90 chunks, and after six action chunks. At each state, eight unmodified runs are used to select one
 91 pair based only on action and endpoint distance. We evaluate both transplant directions with three
 92 future-noise draws each.

93 For Cosmos 3, we intervene in the 36-layer diffusion-generator component of the 16B
 94 `Cosmos3-Nano-Policy-DROID` checkpoint, where future-video and action coordinates are
 95 jointly denoised [14, 15]. We evaluate one state from each of four RoboLab tasks selected with-
 96 out intervention outcomes [19]. These states provide 12 donor continuations. Each donor is tested
 97 with its model-predicted future and with a future encoded from the video produced by executing the
 98 donor action. A fifth Banana state is reserved for the held-out key-value experiment.

99 Controls use the recipient’s own future, Gaussian noise matched to the donor future’s norm and
 100 distance from the recipient, and a within-task shuffled future. No-op recomputations verify that the
 101 intervention code leaves unmodified runs unchanged. At each patched Cosmos 3 attention operation,
 102 we replace only keys and values belonging to future-video tokens; no action coordinate or action-
 103 token key or value is patched directly.

104 The independent unit is a restored simulator state. Reciprocal directions, donors, noise draws, and
 105 process launches are repeated measurements of that state. We average within each state and compute
 106 95% confidence intervals with 10,000 cluster-bootstrap resamples over states [4]. For Cosmos 3,
 107 this is equivalent to resampling the four tasks. The prespecified smallest effect of interest for the
 108 confirmatory Cosmos Policy Predict2 experiments is 0.10 projection units [9].

109 3.3 Content and pathway interventions

110 We use three experiments to identify which changes in a Cosmos Policy future are associated with
 111 the action shift. First, at the ten first-query states, we replace the wrist-camera future, primary-
 112 camera future, or future proprioception separately. Second, at 20 held-out states, we independently
 113 re-render binary robot state R and object state O , producing four 2×2 combinations per state. A
 114 decoder identifies the intended combination in 74/80 primary-camera videos and 79/80 wrist-camera
 115 videos, confirming that the manipulation is visible in both views. The available analysis reports the
 116 object-state main effect, but not the robot-state main effect. Third, we search unmodified policy
 117 continuations for natural pairs that differ primarily in robot motion or object state. Only four states
 118 meet the matching criteria, below the prespecified minimum of ten, so this analysis is exploratory.

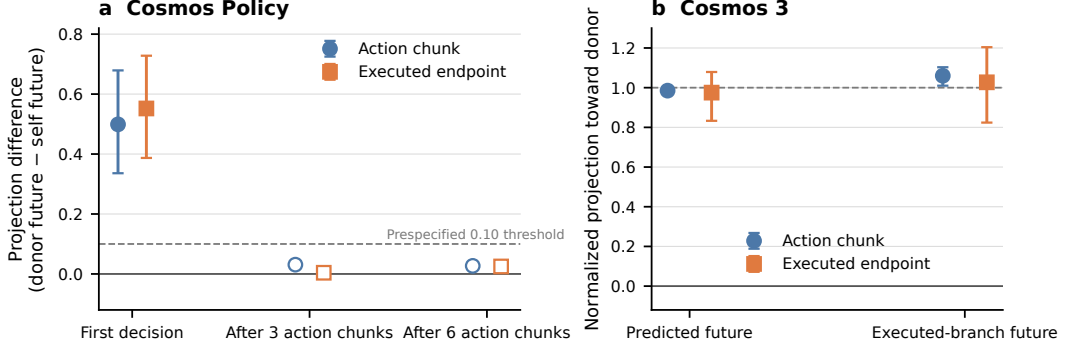


Figure 2: Directional effects of future transplantation. (a) Cosmos Policy reports the donor-minus-self projection difference Δ ; effects are large at first-query states and small at later sampled states. (b) Cosmos 3 reports raw donor projection π ; 0 reproduces the recipient and 1 reproduces the donor. Intervals are 95% cluster-bootstrap confidence intervals where available.

For the Cosmos 3 pathway experiment, we record future-token keys and values from a recipient-future run, then replace the corresponding activations during a donor-future run. We test replacement at layer 16, at every direct future-to-action attention read, and at those direct reads plus future-to-current-video reads that can carry future information to the action indirectly. Four states are used to calibrate these schemes. We then freeze the interventions and apply them unchanged to the held-out Banana state. We report the decrease in donor projection caused by each replacement. This is an intervention effect, not a formal mediation estimate.

4 Results

4.1 Coherent donor futures steer actions toward donor behavior

Cosmos Policy. At the first policy query, donor futures increase action projection by $\Delta_a = 0.499$ (95% CI [0.336, 0.679]) relative to self-future recomputation. The effect is positive in all ten tasks. Executing the recomputed action from the same state yields a corresponding endpoint effect of $\Delta_e = 0.552$ ([0.387, 0.728]), also positive in all ten tasks. Relative to the Gaussian and shuffled-future controls, the action-projection differences are 0.497 and 0.255, and the endpoint differences are 0.549 and 0.277. Confidence intervals are unavailable for these secondary control contrasts.

Effects are much smaller at the later sampled states (Figure 2a). After three action chunks, $\Delta_a = 0.031$ and $\Delta_e = 0.004$; after six chunks, they are 0.027 and 0.025. Because each decision point uses a different saved state, these measurements do not distinguish temporal decay from state-specific variation.

Cosmos 3. The available Cosmos 3 analysis reports raw donor projection π , rather than the donor-minus-self difference Δ used above; the self-future control is near zero. Across the four independent states, model-predicted donor futures yield a mean action projection of 0.985 (task-bootstrap 95% CI [0.974, 0.993]) and endpoint projection of 0.975 ([0.833, 1.079]). Futures encoded from executed donor branches yield action projection 1.060 ([1.010, 1.103]) and endpoint projection 1.027 ([0.824, 1.204]). All 24 interventions—12 donors evaluated with two future sources—have positive donor projection. Multiple donors improve measurement within a state but do not increase the number of independent states.

Neither model’s selected branch pairs show stable differences in binary task success. The evidence therefore concerns donor-directed action shifts and one-chunk simulator endpoints, not closed-loop task completion.

4.2 Cosmos Policy effects track camera-visible robot motion

Replacing one future component at a time reveals a large difference across modalities (Figure 3a). Wrist-camera futures produce a mean action effect of 0.435 ([0.284, 0.605]), primary-camera futures

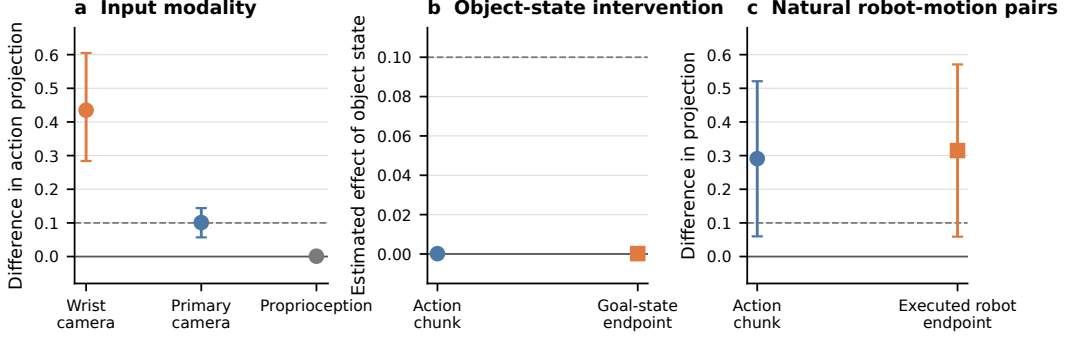


Figure 3: Cosmos Policy content interventions. (a) Action effects from replacing each future modality. (b) Estimated object-state main effect in the factorized intervention. (c) Action and robot-endpoint effects in four exploratory natural robot-motion pairs. Bars show 95% cluster-bootstrap confidence intervals.

152 produce 0.101 ([0.057, 0.144]), and future proprioception produces 0.001 ([−0.004, 0.008]). The
 153 strongest effect therefore comes from the view that most closely tracks the manipulator, while the
 154 proprioception interval includes zero.

155 In the factorized 2×2 experiment, changing object state while averaging over robot state pro-
 156 duces a mean action effect of 0.00015 ([−0.00012, 0.00049]) and endpoint effect of 0.00018
 157 ([−0.00040, 0.00102]). These near-zero effects occur even though the decoder reliably identifies the
 158 rendered conditions. The result isolates the object-state main effect; because the robot-state main
 159 effect was not reported, it does not by itself establish a robot-versus-object contrast. The re-rendered
 160 videos may also lie outside the distribution of futures normally generated by the policy.

161 The four exploratory natural pairs show the same qualitative pattern without re-rendering. Futures
 162 that differ primarily in robot motion produce a mean action effect of 0.291 ([0.060, 0.521]) and robot-
 163 endpoint effect of 0.315 ([0.059, 0.571]). Pairs that differ primarily in object state produce means
 164 of 0.021 for both action and goal-endpoint projection, with intervals that include zero. Because this
 165 analysis contains only four states and incomplete interval reporting for the object pairs, it should be
 166 treated as suggestive.

167 Together, the camera interventions and natural pairs are consistent with sensitivity to camera-visible
 168 robot motion at the sampled Cosmos Policy states. They do not show that the policy generally
 169 ignores object consequences or localize a latent robot-motion representation.

170 4.3 Future-token keys and values contribute to the Cosmos 3 action shift

171 On four calibration states, donor futures produce a mean action projection of 0.981 before key-
 172 value replacement. Replacing future-token keys and values at layer 16 reduces projection by 0.346
 173 units ([0.192, 0.501]). Replacement at all direct future-to-action attention reads reduces projection
 174 by 0.842 ([0.766, 0.910]), with a positive reduction in all four states. Extending the replacement
 175 to future-to-current-video reads reduces projection by 0.995 ([0.976, 1.013]). Record, replay, and
 176 same-run replacement controls reproduce the original actions exactly.

177 The frozen interventions produce the same ordering on the held-out Banana state (Figure 4b). For a
 178 model-predicted donor future, action projection falls from 0.900 to 0.749 after layer-16 replacement,
 179 0.122 after direct replacement, and −0.012 after the extended replacement. For a future encoded
 180 from an executed donor branch, the corresponding values are 1.577, 1.326, 0.382, and −0.012.

181 Endpoint effects are less consistent. For the predicted future, direct replacement changes endpoint
 182 projection from 0.973 to 0.242; for the executed-branch future, it changes endpoint projection only
 183 from 1.019 to 0.976 despite a large action effect. The intervention therefore identifies future-token
 184 attention activations that contribute to the donor-directed action shift, but it does not establish what
 185 they encode or whether they are the only route from future tokens to action.

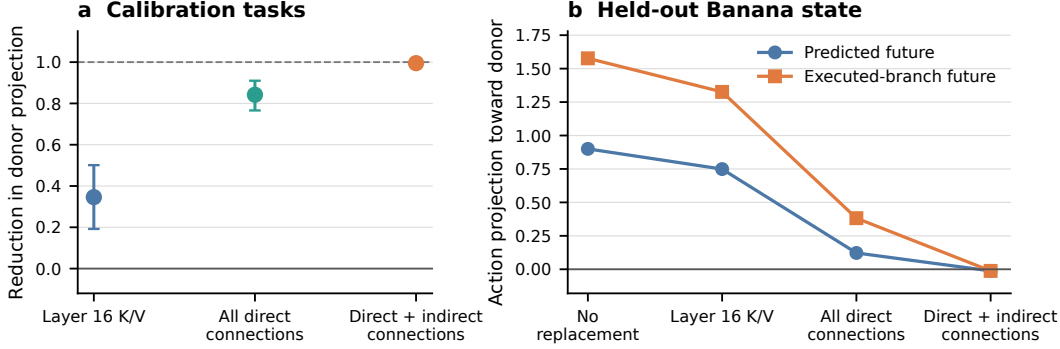


Figure 4: Future-token key-value replacement in Cosmos 3. (a) Reduction in donor-action projection across four calibration states. (b) Action projection on the held-out Banana state under the frozen replacement schemes.

5 Discussion and Limitations

Future transplantation provides stronger evidence than corruption alone because it asks whether an action change follows the content of a specific alternative future. Positive donor projections in both checkpoints show that future representations do more than occupy behaviorally sensitive positions: selecting a different coherent future predictably changes the direction of the action.

In Cosmos Policy, the camera and natural-pair results are consistent with an inverse-dynamics-like use of visible robot motion. The action follows a displayed robot future, whereas future proprioception and isolated object-state changes have near-zero effects. This is not evidence that predicted object consequences never matter. The synthetic intervention tests one restricted object-state contrast, and the natural analysis contains only four states.

The content and pathway results also do not form one mechanism. Cosmos Policy identifies future changes associated with the action shift, while the Cosmos 3 intervention identifies attention activations that contribute to its shift. Because these experiments use different models, we do not show that camera-visible robot-motion information passes through the replaced Cosmos 3 keys and values. More broadly, neither experiment tests candidate generation, outcome comparison, or value-based choice. The results therefore establish future-dependent action generation, not planning.

Limitations. The evidence comes from a limited number of independent states. Cosmos Policy’s largest effects occur at one first-query state from each of ten tasks; later estimates are much smaller, and decision point is confounded with saved state. Cosmos 3 transplantation uses four states, while the held-out key-value test uses one. Additional donors and process launches improve within-state measurement but do not establish generalization across states. We also do not establish effects on task success.

The content and pathway analyses have further limits. Factorized videos may be outside the policy’s normal future distribution and provide only the object-state main effect. The natural matched-pair analysis falls below its prespecified sample size. On the held-out Cosmos 3 state, key-value replacement does not consistently reduce endpoint effects even when it reduces action projection. These qualifications limit the interpretation to the sampled checkpoints and states.

Broader impact. Future transplantation may help audit whether a robot policy’s predictions affect its actions, but the same intervention could redirect behavior. Any use outside simulation would require broader state coverage, physical safety constraints, and independent action monitoring.

6 Conclusion

Future transplantation provides donor-specific evidence that predicted futures influence actions in both WAM checkpoints. Connecting camera-visible robot motion to the future-token key-value pathway within one model, across more states and task outcomes, remains an open problem.

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A Reproducibility Details

The Cosmos Policy experiments use public commit 18a2acca, checkpoint revision cb689ec0, LIBERO, PyTorch 2.7 with CUDA 12.8, and two NVIDIA H100 80GB GPUs. Endpoint vectors concatenate goal-relevant simulator coordinates with the official nine-dimensional proprioceptive endpoint. The prespecified design includes ten tasks, three decision points, eight unmodified policy runs per state, one reciprocal donor pair per state, and three future-noise draws per direction. Confidence intervals use 10,000 bootstrap resamples of simulator states.

The Cosmos 3 experiments use checkpoint revision 2b9f9517. The public report and checkpoint describe a diffusion generator with 36 transformer layers, four diffusion steps, and 32-action chunks of absolute joint positions [14, 15]. For each endpoint evaluation, we reconstruct the simulator, restore the saved state, and replay the same observation prefix. Separate controls check token layout, sampler wrapping, cache recording and replay, and same-run replacement. Sentinel values verify that interventions to future representations never write to the action-token slice directly. Repeating the held-out action intervention in five independently launched processes yields bitwise-identical results; these repetitions test numerical stability, not generalization.

The supplied artifact contains the aggregate values used for the figures, but not the full seed list, intervention logs, code, repository hashes, container digests, per-run wall-clock times, or aggregate compute. These items must be added to a complete reproduction package.

B Additional Results and Exclusions

Across all 30 Cosmos Policy states, the mean donor-minus-self effect is 0.186 for actions and 0.194 for endpoints. Because the three decision points may be nested within task trajectories, the first-query analysis is the cleaner confirmatory result. For Cosmos 3, action-space nearest-donor accuracy

324 is 12/12 for both future sources. Endpoint nearest-donor accuracy is 10/12 for predicted futures and
325 11/12 for executed-branch futures.

326 We separately evaluated six states in RoboCasa [13]. This exploratory experiment is not pooled with
327 LIBERO. The donor-minus-self endpoint effect is 0.0287 (95% CI [0.0101, 0.0492]) and is positive
328 in all six states. The action effect is smaller and does not reliably exceed the Gaussian control. An
329 unfinished 20-state Cosmos 3 study is excluded from all reported estimates.

330 The original draft also reported a natural control whose construction was not documented and a
331 Cosmos Policy attention ablation whose gating rule, independent-state count, confidence intervals,
332 and endpoint effects were unavailable. We exclude both from the revised paper rather than treat
333 them as interpretable evidence.

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